Adaptive Control for Uncertain Nonlinear Systems Against DoS Attacks Using Quantized Output Only and Application to a Single-Link Robot

Shenghang Liu, Xinjun Wang, Wendi Chen, Ben Niu, Xinmin Song, Huanqing Wang, and Xudong Zhao

***Abstract*— This paper investigates the problem of adaptive output feedback control for uncertain nonlinear systems with input/output quantization subject to intermittent denial-of-service (DoS) attacks. When a DoS attack becomes active, the system is unable to obtain output signal. In order to address this challenge, we compensate the output signal by introducing an attack compensator. At the same time, a novel quantization compensator-based (QC) state observer is designed, which uses only the quantized compensated signal to reconstruct the system states. In addition, the problem of over-parameterisation is avoided by using parameter projection technique to design the parameter estimator. By combining Lyapunov stability theory and a modified average dwell time (ADT) approach, the stability of the closed-loop system is ensured. Furthermore, the proposed controller ensures that all closed-loop signals are bounded, and the control performance of mean-square error can be adjusted by appropriately selecting certain design parameters. Finally, the effectiveness of the proposed control strategy is validated through a single-link robot simulation results.**

***Note to Practitioners*—Considering that resources are limited, the DoS attacks mentioned in this paper are aperiodic, which means that attackers engage in frequent attacks within a relatively short time frame. This aligns more closely with real-world scenarios of network attacks. When the attack is active, the output signal is not available for the system. In this paper, a new QC state observer is designed, which in turn enables the estimation of the system state signals. Furthermore, this paper considers both input and output quantization, which reduces duplicate transmission of samples and improves the use of communication resources. This makes the work in this paper more relevant to the practical engineering context.**

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***Index Terms*— Nonlinear systems, adaptive control, attack compensator, DoS attacks, output quantization.**

I. Introduction

In RECENT years, with the rapid advancement of the network, there has been extensive scholarly attention directed towards the exploration of how to use the network for remote control of the system more conveniently and accurately. The network control system is a real-time feedback control system in which each control component is interconnected through the network, ultimately forming a closed-loop control system. All control devices within the system are regarded as network nodes, and data exchange among all nodes occurs through the network. Nowadays, the practical network control systems face various malicious cyber-attacks, such as denial-of-service (DoS) attacks [1], [2], and false data injection attacks [3], [4]. When DoS attacks occur, the network communication channels may be interrupted or blocked, which can result in data not being transmitted through the channels properly [5]. False data injection attacks are achieved by maliciously altering the communication signals, causing the system to be unable to obtain accurate data [6]. It is worth noting that DoS attacks are the most representative form of cyber-attacks, which can cause extremely detrimental effects on the performance and stability of the systems. Therefore, the security control problems of network control systems subject to DoS attacks have attracted the attention of scholars. The robust control problems of linear systems under DoS attacks were investigated in [7]. A new resilient triggering control strategy under DoS attacks was proposed in [8]. For wireless network control systems, a resilient sampling control scheme against DoS attacks was designed in [9]. For linear Cyber-Physical Systems under DoS attacks, an output feedback strategy was proposed to achieve the control objective of cooperative tracking [10]. It should be noted that the above literatures are primarily based on studies involving linear systems. However, when dealing with real industrial systems, various nonlinear factors are often present. This urges us to conduct more in-depth and more precise research on nonlinear systems. Under DoS attacks [11], a fault-tolerant cooperative control framework was constructed based on a switching-type observer. For nonlinear network control systems subjected to DoS attacks, a resilient control strategy was proposed [12]. In [13], a class of multiagent systems were considered, and

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an event-triggered security control scheme was proposed to address DoS attacks. A novel adaptive memory mechanism was introduced in [14] to further improve the observer accuracy for nonlinear multiagent systems under DoS attacks.

In practical network control systems, the channel bandwidth is limited. To ensure that the systems operate within the given bandwidth constraints, it is necessary to employ quantization techniques to reduce the communication rate. Despite the benefits associated with the introduction of this technology, this can also lead to a certain amount of data loss, thus reducing the performance of the system. Therefore, the network control systems with quantization need to be studied in depth to ensure that their performance remains robust and effective. In [15] and [16], based on robust methods, the problems of quantization control of uncertain nonlinear systems were considered. For nonlinear discrete-time systems, a quantized feedback control method was investigated [17]. By introducing indirect fractional-order Lyapunov stability, a modified backstepping control method was developed for fractional-order nonlinear systems with input quantization [18]. In [19], the stability problem of strict feedback nonlinear systems with input quantization was investigated based on the selection criteria for quantizer parameters. However, the above-mentioned schemes only consider the quantization of input signals, while in practical applications, both input and output signals need to be considered. Some progress in robust control regarding quantized states and outputs were presented in [20] and [21]. A model reference control scheme was constructed for general discrete-time linear time-invariant systems with output quantization and saturation in [22]. The adaptive control schemes for nonlinear systems with both input and output quantization were investigated in [23] and [24]. In order to ensure the stability of uncertain nonlinear systems with state quantization, a control scheme based on adaptive backstepping was first introduced in [25], and this approach was extended to systems with unmatched nonlinearities by [26]. In [27], the tracking performance was guaranteed for uncertain nonlinear systems in the presence of input/output quantization. The research findings indicated that the proposed controller not only overcame the challenges of asymmetric actuator saturation but also ensured the semi-global practical finite-time stability of the closed-loop system. Note that the above research results don't consider the impacts of cyber-attacks. However, the network control systems with input/output quantization are often vulnerable to cyber-attacks, which motivates the work in this paper.

In this paper, we study the control problem of nonlinear systems with input/output quantization under DoS attacks. The main contributions of this paper are as follows.

1) This paper considers a more general class of uncertain nonlinear systems with mismatched parameters, where the estimated parameters and the partial derivatives of the virtual signals with respect to the estimated parameters as well as the state variables are nonlinear and time-varying. In order to overcome this challenge, the paper introduces the concept of dynamic filtering technology to construct a first-order filter, which enables the controller design to avoid differentiation of virtual control signals directly.
2) Unlike in references [25] and [26], we incorporate projection operations into the parameter adaptive law, ensuring both the boundedness of the parameter estimate and effective handling of the effects introduced by output quantization. This approach saves computational costs and avoids the impact of over-parameterization. Meanwhile, compared to the existing literatures [19], [28], and [29], this paper distinguishes itself by considering both input and output quantization, which enhances the practical engineering applicability of the research content.
3) When a DoS attack becomes active, the system is unable to acquire output signal. This paper introduces an attack compensator to compensate for the output signal, thereby proposing a novel quantization compensator-based (QC) state observer. The state observer utilizes the quantized compensation signal to estimate the system states while simultaneously ensuring that the system can still achieve its objective of security control under the condition of attack. In addition, in order to ensure the stability of the system, an improved average dwell time (ADT) method is proposed by combining the traditional average dwell time methods [30] and [31] with the frequency and duration of DoS attacks, thus realizing the expansion from periodic to aperiodic.

The remaining organization of this paper is as follows. Section II describes the system model and outlines some preliminary work. In Section III, an adaptive security control scheme is proposed, with stability analysis provided at the end of the section. Section IV presents illustrative simulation results and we conclude the paper in Section V.

Notation: $\lfloor\star\rfloor$ denotes the greatest integer that is less than or equal to $\star$. I is the identity matrix.

II. Mathematical Preliminaries and Problem Formulation

A. Quantizer

The various quantizers referred in this paper all satisfy the following property:

$$|q(s) - s| \le \xi_s, \tag{1}$$

where the quantization error bound is defined as ξ_s with $\xi_s > 0$. The uniform quantizer and logarithmic-uniform quantizer are shown in detail below, respectively.

1) Uniform Quantizer: The specific construction of the uniform quantizer is given in the following equation:

$$q_{uni}(s) = \begin{cases} \mu_i sgn(s), & \mu_i - \dfrac{z}{2} < |s| \le \mu_i + \dfrac{z}{2}, \\ 0, & |s| \le \mu_0, \end{cases} \tag{2}$$

where $\mu_0 = \frac{z}{2}$, $\mu_1 = \mu_0 + \frac{z}{2}$, $\mu_{i+1} = \mu_i + z$ with $i = 1, \ldots,$ the size of the deadzone for $q_{uni}(s)$ is determined by μ_0, and $q_{uni}(s) \in \mathfrak{U} = \{0, \pm\mu_i\}$. The length of the quantization interval is denoted by the parameter z with z>0. The uniform quantizer satisfies property (1) with $\xi_s = \max\{\mu_0, \frac{z}{2}\}$.

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2) Logarithmic-Uniform Quantizer: Based on the logarithmic-uniform quantizer proposed in [28], we further describe it as

$$q_{lu}(s)=\begin{cases} q_{log}(\mu_{lu})+\lfloor\dfrac{s-\mu_{lu}}{\varrho}+\theta\rfloor\varrho, & |s|\geq\mu_{lu},\\ q_{log}(s), & |s|<\mu_{lu}, \end{cases} \tag{3}$$

where μ_{lu} represents the threshold to switch between the logarithmic and uniform quantizer with μ_{lu} being a given positive constant. ϱ is a parameter of the uniform quantizer and satisfies $\varrho=|q_{log}(\mu_{lu})-\mu_{lu}|$. For the value of the parameter θ in (3) is described by the following rules: $\theta=1$ if $q_{log}(\mu_{lu})<\mu_{lu}$, $\theta=0$ if $q_{log}(\mu_{lu})\geq\mu_{lu}$. $q_{log}(s)$ denotes the output of a logarithmic quantizer and the specific construction is described as the following form:

$$q_{log}(s)=\begin{cases} \mu_i sgn(s), & \dfrac{\mu_i}{1+m}<|s|\leq\dfrac{\mu_i}{1-m},\\ 0, & |s|\leq\dfrac{\mu_{min}}{1+m}, \end{cases} \tag{4}$$

where $\mu_i=k^{(1-i)}\mu_{min}, i=1,2,\ldots$ and $k=\frac{1-m}{1+m}$ with $0<m<1$. The size of the deadzone for $q_{log}(s)$ is determined by $\frac{\mu_{min}}{1+m}$, which satisfies $\frac{\mu_{min}}{1+m}>0$. The uniform quantizer also satisfies property (1), where $\xi_s=\max\{\frac{m-m^2}{1+m}\mu_{lu},\varrho\}$ is the maximum quantization interval length.

Remark 1: It is worth noting that the value of the parameter μ_{lu} is determined by the user based on the actual situation. It is used to regulate the relationship between system performance and communication burden.

B. Intermittent DoS Attacks

The rapid development of the network is also accompanied by a variety of cyber attacks. In this paper, we consider adversaries maliciously conducting intermittent DoS attacks on the communication channel, which means that the attacks are discontinuous. Denote the moments at the begins and ends of the lth DoS attack by T_l^a and T_l^b, respectively. Then, we can get the lth DoS attack interval $T_l:=[T_l^a,T_l^b)$. During this time interval, the DoS attack is active and data cannot be successfully transmitted through the communication channel.

Combine the above content, the sum of the time for the DoS attacks that occur in the interval $[\pi,t]$ can be defined as

$$\Theta_a(\pi,t)=\bigcup_{l\in N}T_l\bigcap[\pi,t], \tag{5}$$

where $N>0$ is an integer, $0\leq\pi$ and $0\leq t$ with $t\geq\pi$.

Then, the sum of attack sleep times can be defined as

$$\Theta_b(\pi,t)=[\pi,t]\backslash\Theta_a(\pi,t). \tag{6}$$

Assumption 1 (DoS Duration [32]): There exists a constant $\mathcal{T}\geq 1$ and satisfies

$$|\Theta_a(\pi,t)|\leq\frac{t-\pi}{\mathcal{T}}. \tag{7}$$

Assumption 2 (DoS Frequency [32]): There exist constants $\varkappa>0$ and $\tau_D>0$ satisfying

$$n(\pi,t)\leq\varkappa+\frac{t-\pi}{\tau_D}. \tag{8}$$

Remark 2: In reality, due to considerations of limited resources, attackers typically avoid adopting periodic attack patterns. They prefer launching frequent attacks over shorter durations, which means that the attack methods employed are aperiodic. This paper primarily focuses on the aperiodic DoS attacks. Therefore, the constraints imposed on the duration and frequency of DoS attacks based on Assumptions 1 and 2 are reasonable. The above assumptions can be found in [32] and [33].

C. System Descriptions

This paper studies the uncertain nonlinear system with input/output quantization, which is described as follows:

$$\begin{cases} \dot{x}_i=x_{i+1}+f_i(y)+\Psi^T\kappa_i(y),\ i=1,\ldots,n-1,\\ \dot{x}_n=q(u)+f_n(y)+\Psi^T\kappa_n(y),\\ y=x_1, \end{cases} \tag{9}$$

where $x_i\in\Re(i=1,\ldots,n)$ and $y\in\Re$ are the state variables and output of the system, respectively. $u\in\Re$ is the designed controller. $q(u)$ is the quantized value obtained from the quantizer, where the quantization error satisfies $|q(u)-u|\leq\xi_u$. $f_i(y)$ and $\kappa_i(y)\in\Re^r$ are the known and smooth nonlinear functions, where $|f_i(y)|\leq\bar{f}$ and $||\kappa_i(y)||\leq\bar{\kappa}$ with $\bar{f}$ and $\bar{\kappa}$ being unknown constants. $\Psi\in\Re^r$ is a vector of unknown parameters and satisfies $\Psi\in\Omega_\Psi:=\{\Psi:\|\Psi\|\leq\Psi_0\}$, where Ω_Ψ is a known compact convex set and Ψ_0 is a positive constant.

For the uncertain nonlinear system (9) with input/output quantization under DoS attacks, the main objective of this paper is to design an adaptive output feedback control scheme, which ensures the stability of the closed-loop system and the boundedness of all signals.

Assumption 3: For functions $f_i(y)$ and $\kappa_i(y), i=1,\ldots,n$, satisfy the following global Lipschitz continuity conditions:

$$\|\kappa_i(y_1)-\kappa_i(y_2)\|\leq\mathcal{L}_{\kappa_i}|y_1-y_2|, \tag{10}$$

$$|f_i(y_1)-f_i(y_2)|\leq\mathcal{L}_{f_i}|y_1-y_2|, \tag{11}$$

where $\mathcal{L}_{\kappa_i}$ and $\mathcal{L}_{f_i}$ are known Lipschitz constants.

III. Main Results

In this section, an adaptive control strategy based on the QC observer is proposed to counter malicious DoS attacks for the nonlinear system (9), whose logical control framework is shown in Fig. 1.

A. Quantization Compensator-Based State Observer Design

Let $\check{y}$ denote the actual measurement output when system (9) is attacked, then, the attack compensator can be designed as

$$\check{y}=\begin{cases} x_1(T_l^{a-}), & t\in[T_l^a,T_l^b),\\ x_1(t), & t\in[T_l^b,T_{l+1}^a), \end{cases} \tag{12}$$

where T_l^{a-} is the left limit of T_l^a. Define $y^a=x_1(T_l^{a-})$ and $y^b=x_1(t)$ with $x_1(T_l^{a-})$ being the last obtained output signal before the lth DoS attack occurs.

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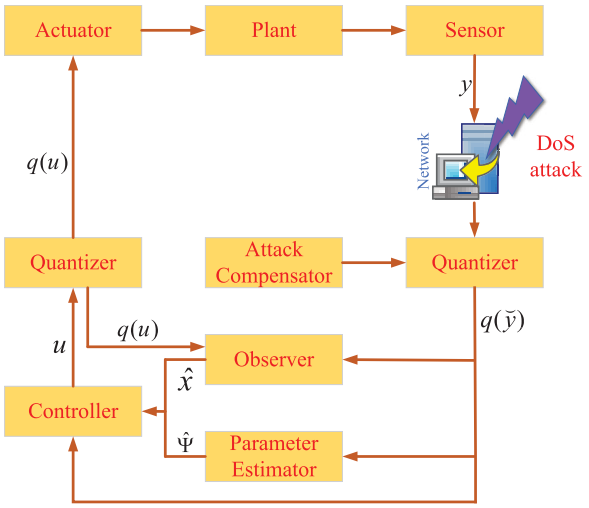


Fig. 1. Logical framework of the proposed control strategy.

In order to estimate state signals x_i with $i = 2, \ldots, n$ that cannot be measured under DoS attacks via the quantized compensation output, a novel QC observer is presented

$$\begin{cases} \dot{\hat{x}}_i = \hat{x}_{i+1} + f_i(q(\check{y})) + \hat{\Psi}^T \kappa_i(q(\check{y})) + \ell_i(q(\check{y}) - \hat{y}), \\ \dot{\hat{x}}_n = q(u) + f_n(q(\check{y})) + \hat{\Psi}^T \kappa_n(q(\check{y})) + \ell_n(q(\check{y}) - \hat{y}), \\ \hat{y} = \hat{x}_1, \ i = 1, \ldots, n-1, \end{cases} \tag{13}$$

where $\hat{x}_i (i = 1, \ldots, n)$ is the estimate of x_i, $\hat{\Psi}$ is the estimate of Ψ. When DoS attacks occur, $\hat{\Psi}$ is restated as $\hat{\Psi}_a$; otherwise, $\hat{\Psi}$ is represented as $\hat{\Psi}_b$. $q(\check{y})$ is the quantized compensation output signal. From the property (1) of the quantizer, the quantization error satisfies

$$|q(\check{y}) - \check{y}| \le \xi_{\check{y}}, \tag{14}$$

where $\xi_{\check{y}} > 0$ is the quantization bound.

Let $e_i = x_i - \hat{x}_i$ be an estimate error with $i = 1, \ldots, n$. According to (9) and (13), we can obtain that the error e satisfies

$$\dot{e} = \begin{cases} A_1 e - \mathfrak{L}_\mathfrak{a} M \bar{e}_1^{\,a} + f(y) - f(q(y^a)) + \Psi_a^T \kappa(y) \\ -\hat{\Psi}_a^{\,T} \kappa(q(y^a)) + \mathfrak{L}_\mathfrak{a}(y^a - q(y^a)), t \in [T_l^a, T_l^b), \\ A_2 e + f(y) - f(q(y^b)) + \Psi_b^T \kappa(y) - \hat{\Psi}_b^{\,T} \kappa(q(y^b)) \\ +\mathfrak{L}_\mathfrak{b}(y^b - q(y^b)), \ t \in [T_l^b, T_{l+1}^a), \end{cases} \tag{15}$$

where $e = [e_1, e_2, \ldots, e_n]^T$ and we define $\bar{e}_1^a = y^a - \hat{y}$. $\kappa(y) = [\kappa_1(y), \ldots, \kappa_n(y)]^T$, $f(y) = [f_1(y), \ldots, f_n(y)]^T$, $\mathfrak{L}_\mathfrak{a} = [\ell_1^a, \ldots, \ell_n^a]^T$ and $\mathfrak{L}_\mathfrak{b} = [\ell_1^b, \ldots, \ell_n^b]^T$. $A_2 = A_1 - \mathfrak{L}_\mathfrak{b} M$ with

$$A_1 = \begin{bmatrix} 0 & 1 & & & \\ 0 & 0 & 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & \ldots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix}, \quad M = \begin{bmatrix} 1, 0, \cdots 0 \end{bmatrix}^T. \tag{16}$$

For $t \in [T_l^a, T_l^b)$, we can express the definition of e_1^a as follows: $e_1^a = q(y^a) - \hat{y}$. Then, the Lyapunov function can be selected as

$$V_e^a = e^T \mathcal{P}_a e, \tag{17}$$

where $\mathcal{P}_a > 0$ is a symmetric matrix, the derivative of V_e^a is given as

$$\begin{aligned} \dot{V}_e^a = {} & e^T (\mathcal{P}_a A_1 + A_1^T \mathcal{P}_a) e + 2 e \mathcal{P}_a \big[f(y) - f(q(y^a)) \\ & - \mathfrak{L}_\mathfrak{a} M \bar{e}_1^a + \Psi_a^T \kappa(y) - \hat{\Psi}_a^T \kappa(q(y^a)) \\ & + \mathfrak{L}_\mathfrak{a}(y^a - q(y^a)) \big]. \end{aligned} \tag{18}$$

Using Young's inequality, the following inequality can be obtained:

$$\begin{aligned} \dot{V}_e^a \le {} & e^T (\mathcal{P}_a A_1 + A_1^T \mathcal{P}_a) e + 2\|e\| \|\mathcal{P}_a\| \big\| \mathcal{L}_f^a \big\| \big| y - q(y^a) \big| \\ & + 2 e \mathcal{P}_a (\kappa^T(y) \Psi_a - \kappa^T(q(y^a)) \Psi_a + \kappa^T(q(y^a)) \tilde{\Psi}_a) \\ & + 2\|e\| \|\mathcal{P}_a\| \|\mathfrak{L}_\mathfrak{a}\| \xi_{\check{y}} - 2 e^T \mathcal{P}_a \mathfrak{L}_\mathfrak{a} M \bar{e}_1^a \\ \le {} & e^T (\mathcal{P}_a A_1 + A_1^T \mathcal{P}_a) e + 2\|e\| \|\mathcal{P}_a\| \big\| \mathcal{L}_f^a \big\| (|y| + \big|q(y^a)\big|) \\ & + 2\|e\| \|\mathcal{P}_a\| \|\mathfrak{L}_\mathfrak{a}\| \xi_{\check{y}} + (\frac{1}{c} + c \|\mathcal{P}_a \mathfrak{L}_\mathfrak{a} M\|^2) \|e\|^2 \\ & + 2\|e\| \|\mathcal{P}_a\| \big\| \mathcal{L}_\kappa^a \big\| \|\Psi_a\| (|y| + \big|q(y^a)\big|) \\ & + 2\|e\| \|\mathcal{P}_a\| \big\| \mathcal{L}_\kappa^a \big\| \big|q(y^a)\big| \Psi_m \\ \le {} & e^T (\mathcal{P}_a A_1 + A_1^T \mathcal{P}_a + \frac{2}{c} + c \|\mathcal{P}_a \mathfrak{L}_\mathfrak{a} M\|^2 + \frac{4}{c} \big\| \mathcal{P}_a \|^2 \|\mathcal{L}_f^a \big\|^2 \\ & + \frac{4}{c} \big\| \mathcal{P}_a \|^2 \|\mathcal{L}_\kappa^a \big\|^2 \Psi_0^2) e + \frac{1}{2} c y^2 + \delta_a^0, \end{aligned} \tag{19}$$

where c is a positive constant, $\delta_a^0 = 4c\|\mathcal{P}_a\|^2 \|\mathcal{L}_f^a\|^2 q^2(y^a) + 4c\Psi_0^2 \|\mathcal{P}_a\|^2 \|\mathcal{L}_\kappa^a\|^2 q^2(y^a) + 4c\Psi_m^2 \|\mathcal{P}_a\|^2 \|\mathcal{L}_\kappa^a\|^2 q^2(y^a) + 4c\|\mathcal{P}_a\|^2 \|\mathfrak{L}_\mathfrak{a}\|^2 \xi_{\check{y}}^2$. In addition, $\mathcal{L}_\kappa^a = [\mathcal{L}_{\kappa_1}^a, \ldots, \mathcal{L}_{\kappa_n}^a]^T$, $\mathcal{L}_f^a = [\mathcal{L}_{f_1}^a, \ldots, \mathcal{L}_{f_n}^a]^T$. Ψ_m will be defined in the following section.

For $t \in [T_l^b, T_{l+1}^a)$, let $e_1^b = q(y^b) - \hat{y}$. Choose the Lyapunov function as

$$V_e^b = e^T \mathcal{P}_b e, \tag{20}$$

where $\mathcal{P}_b > 0$ is symmetric matrix. The derivative of V_e^b is given as

$$\begin{aligned} \dot{V}_e^b = {} & e^T (\mathcal{P}_b A_2 + A_2^T \mathcal{P}_b) e + 2 e \mathcal{P}_b \big[f(y) - f(q(y^b)) \\ & + \Psi_b^T \kappa(y) - \hat{\Psi}_b^T \kappa(q(y^b)) + \mathfrak{L}_\mathfrak{b}(y^b - q(y^b)) \big]. \end{aligned} \tag{21}$$

Similar to (19), the following inequality holds:

$$\dot{V}_e^b \le e^T (\mathcal{P}_b A_2 + A_2^T \mathcal{P}_b + \frac{1}{c}) e + \delta_b^0, \tag{22}$$

where c is a positive constant. $\delta_b^0 = 4c\|\mathcal{P}_b\|^2 \|\mathcal{L}_f^b\|^2 \xi_{\check{y}}^2 + 4c\Psi_0^2 \|\mathcal{P}_b\|^2 \|\mathcal{L}_\kappa^b\|^2 \xi_{\check{y}}^2 + 4c\|\mathcal{P}_b\|^2 \|\mathfrak{L}_\mathfrak{b}\|^2 \xi_{\check{y}}^2 + 4c\Psi_m^2 \|\mathcal{P}_b\|^2 \|\mathcal{L}_\kappa^b\|^2 \xi_{\check{y}}^2$. Besides, $\mathcal{L}_\kappa^b = [\mathcal{L}_{\kappa_1}^b, \ldots, \mathcal{L}_{\kappa_n}^b]^T$, $\mathcal{L}_f^b = [\mathcal{L}_{f_1}^b, \ldots, \mathcal{L}_{f_n}^b]^T$. Ψ_m will be defined in the following section.

B. Security Controller Design

In this subsection, we construct an observer-based safety controller for nonlinear system (9) by combining dynamic filtering technique as well as backstepping design method.

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Furthermore, we will prove the effectiveness of this controller in the section on stability analysis.

According to the observer (13) and the first-order low-pass filter, the following coordinate transformations can be obtained:

$$\begin{aligned}\zeta_1 &= x_1,\\ \zeta_k &= \hat{x}_k - \vartheta_k, \quad k = 2, \ldots, n,\end{aligned} \tag{23}$$

where ζ_k $(k = 1, 2, \ldots, n)$ is the error variable, and ϑ_k is the output of the following first-order low-pass filter with respect to the virtual controller α_{k-1}:

$$\varpi_k \dot{\vartheta}_k + \vartheta_k = \alpha_{k-1}, \vartheta_k(0) = \alpha_{k-1}(0), \quad k = 2, \ldots, n, \tag{24}$$

where ϖ_k is the positive design constant. In addition, we define the filter error as

$$\varepsilon_k = \vartheta_k - \alpha_{k-1}, \quad k = 2, \ldots, n. \tag{25}$$

From the the standard parameter estimation algorithm defined in [34] and [35], we can design the parameter adaptive law as

$$\dot{\hat{\Psi}} = Proj[\Gamma \check{\zeta}_1 \kappa_1(q(\check{y}))], \quad \hat{\Psi}(0) \in \Omega_\Psi, \tag{26}$$

where $\check{\zeta}_1 = q(\check{y})$, Γ is a positive definite design matrix. The projection operator is represented as $Proj(*)$, which is used to ensure the parameter Ψ being estimated is always within the known set $\|\hat{\Psi}\| \le \Psi_c + \iota \triangleq \Psi^*, \forall t \ge 0$ if $\hat{\Psi}(0) \in \Omega_\Psi$. ι denotes an arbitrary constant. Therefore, we can get $\|\tilde{\Psi}\| \le 2\Psi_c + \iota \triangleq \Psi_m$.

Next, the detailed explanation of the design processes will be presented.

Case 1: When $t \in [T_l^a, T_l^b)$, the system is under the malicious Dos attack.

Step 1: Construct the Lyapunov function V_1^a as

$$V_1^a = \frac{1}{2}(\zeta_1^a)^2 + \frac{1}{2}\tilde{\Psi}_a^T \Gamma^{-1} \tilde{\Psi}_a. \tag{27}$$

Differentiating V_1^a, one has

$$\begin{aligned}\dot{V}_1^a = \zeta_1^a \big(e_2 + \zeta_2^a + \varepsilon_2^a + \alpha_1^a + f_1(y) + \Psi_a^T \kappa_1(y)\big)\\ - \tilde{\Psi}_a^T \Gamma^{-1} \dot{\hat{\Psi}}_a.\end{aligned} \tag{28}$$

By using Young's inequality, we can get the following formula:

$$\begin{aligned}\dot{V}_1^a &\le \zeta_1^a \big(\alpha_1^a + f_1(y) + \Psi_a^T \kappa_1(y)\big) + 3(\zeta_1^a)^2\\ &\quad + \frac{1}{4}\big(e_2^2 + (\varepsilon_2^a)^2 + (\zeta_2^a)^2\big) - \tilde{\Psi}_a^T \Gamma^{-1} \dot{\hat{\Psi}}_a\\ &\le \zeta_1^a (\alpha_1^a + f_1(q(y^a)) + \tilde{\Psi}_a^T \kappa_1(y) + \hat{\Psi}_a^T \kappa_1(q(y^a)))\\ &\quad + \frac{1}{4}(e_2^2 + (\varepsilon_2^a)^2 + (\zeta_2^a)^2) + \big|\zeta_1^a\big|\mathcal{L}_{f_1}^a \big(|y| + \big|q(y^a)\big|\big)\\ &\quad + \big|\zeta_1^a\big|\Psi_0 \mathcal{L}_{\kappa_1}^a \big(|y| + \big|q(y^a)\big|\big) + 3(\zeta_1^a)^2\\ &\quad - \tilde{\Psi}_a^T \Gamma^{-1} \dot{\hat{\Psi}}_a\\ &\le \zeta_1^a (\alpha_1^a + \tilde{\Psi}_a^T \kappa_1(y)) + \big|\zeta_1^a\big|\mathcal{L}_{f_1}^a \big(|y| + \big|q(y^a)\big|\big)\\ &\quad + \frac{1}{4}(e_2^2 + (\varepsilon_2^a)^2 + (\zeta_2^a)^2) + \frac{1}{2}(\bar{f}^2 + \Psi_0^2 \bar{\kappa}^2)\\ &\quad + \big|\zeta_1^a\big|\Psi_0 \mathcal{L}_{\kappa_1}^a \big(|y| + \big|q(y^a)\big|\big) + 3(\zeta_1^a)^2\end{aligned}$$

$$\quad - \tilde{\Psi}_a^T \Gamma^{-1} \dot{\hat{\Psi}}_a. \tag{29}$$

The virtual controller α_1^a is designed as

$$\alpha_1^a = -\beta_1^a \hat{x}_1, \tag{30}$$

where β_1^a is a positive constant.

Using Young's inequality, we can get $-\beta_1^a \hat{x}_1 \le -(\beta_1^a/2)(\zeta_1^a)^2 + (\beta_1^a/2)e_1^2$ and substituting (30) into (29) results in

$$\begin{aligned}\dot{V}_1^a &\le (4 + \mathcal{L}_{f_1}^a + \Psi_0 \mathcal{L}_{\kappa_1}^a - \frac{\beta_1^a}{2})(\zeta_1^a)^2 + \frac{\beta_1^a}{2} e_1^2 + \delta_a^1\\ &\quad + \frac{1}{4}(e_2^2 + (\varepsilon_2^a)^2 + (\zeta_2^a)^2) + \tilde{\Psi}_a{}^T \big(\zeta_1^a \kappa_1(y) - \Gamma^{-1} \dot{\hat{\Psi}}_a\big),\end{aligned} \tag{31}$$

where $\delta_a^1 = \frac{1}{2}\mathcal{L}_{f_1}^{a2} q^2(y^a) + \frac{1}{2}\Psi_0{}^2 \mathcal{L}_{\kappa_1}^{a2} q^2(y^a) + \frac{1}{2}(\bar{f}^2 + \Psi_0^2 \bar{\kappa}^2)$.

Then, we use the parametric adaptive law given in (26) as well as combining the property of the projection operator: $-\tilde{\Psi}^T \Gamma^{-1} Proj(\circledast) \le -\tilde{\Psi}^T \Gamma^{-1} \circledast$, the following formula holds:

$$\begin{aligned}&\tilde{\Psi}_a^T (\zeta_1^a \kappa_1(y) - \Gamma^{-1} \dot{\hat{\Psi}}_a)\\ &= \tilde{\Psi}_a^T (\zeta_1^a \kappa_1(y) - \zeta_1^a \kappa_1(q(y^a)) + \zeta_1^a \kappa_1(q(y^a)) - \check{\zeta}_1 \kappa_1(q(y^a))\\ &\quad + \check{\zeta}_1 \kappa_1(q(y^a)) - \Gamma^{-1} Proj(\Gamma \check{\zeta}_1 \kappa_1(q(y^a))))\\ &\le \tilde{\Psi}_a^T (\zeta_1^a \kappa_1(y) - \zeta_1^a \kappa_1(q(y^a)) + \zeta_1^a \kappa_1(q(y^a)) - \check{\zeta}_1 \kappa_1(q(y^a)))\\ &\le \|\tilde{\Psi}_a\| \mathcal{L}_{\kappa_1}^a (|y| + |q(y^a)|)^2 \le (\frac{1}{2}c + \Psi_m \mathcal{L}_{\kappa_1}^a)(\zeta_1^a)^2\\ &\quad + \frac{8}{c}\Psi_m^2 \mathcal{L}_{\kappa_1}^{a2} q(y^a)^2 + \Psi_m \mathcal{L}_{\kappa_1}^a q(y^a),\end{aligned} \tag{32}$$

where c is a positive constant to be designed.

Substituting (32) into (31), it follows that

$$\begin{aligned}\dot{V}_1^a &\le (4 + \mathcal{L}_{f_1}^a + \Psi_0 \mathcal{L}_{\kappa_1}^a + \frac{1}{2}c + \Psi_m \mathcal{L}_{\kappa_1}^a - \frac{\beta_1^a}{2})(\zeta_1^a)^2\\ &\quad + \frac{\beta_1^a}{2} e_1^2 + \delta_a^2 + \frac{1}{4}(e_2^2 + (\varepsilon_2^a)^2 + (\zeta_2^a)^2),\end{aligned} \tag{33}$$

where $\delta_a^2 = \delta_a^1 + \frac{8}{c}\Psi_m^2 \mathcal{L}_{\kappa_1}^{a2} q(y^a)^2 + \Psi_m \mathcal{L}_{\kappa_1}^a q(y^a)^2$.

Step 2: Choose the Lyapunov candidate function as

$$V_2^a = \frac{1}{2}(\zeta_2^a)^2 + \frac{1}{2}(\varepsilon_2^a)^2. \tag{34}$$

Then, the time derivative of V_2^a satisfies

$$\begin{aligned}\dot{V}_2^a &= \zeta_2^a (\zeta_3^a + \alpha_2^a + \varepsilon_3^a + f_2(q(y^a)) + \hat{\Psi}_a \kappa_2(q(y^a))\\ &\quad + \ell_2^a ((q(y^a)) - \hat{y}) - \dot{\vartheta}_2^a) + \varepsilon_2^a \dot{\varepsilon}_2^a.\end{aligned} \tag{35}$$

According to Young's inequality and (24), we have

$$\zeta_2^a \zeta_3^a \le \frac{1}{2}(\zeta_2^a)^2 + \frac{1}{2}(\zeta_3^a)^2, \tag{36}$$

$$\zeta_2^a \varepsilon_3^a \le \frac{1}{2}(\zeta_2^a)^2 + \frac{1}{2}(\varepsilon_3^a)^2, \tag{37}$$

$$\zeta_2^a f_2(q(y^a)) \le \frac{1}{2}(\zeta_2^a)^2 + \frac{1}{2}\bar{f}^2, \tag{38}$$

$$\zeta_2^a \hat{\Psi}_a \kappa_2(q(y^a)) \le \frac{1}{2}(\zeta_2^a)^2 + \frac{1}{2}\Psi_0^2 \bar{\kappa}^2, \tag{39}$$

and

$$\varepsilon_2^a \dot{\varepsilon}_2^a \le -\frac{(\varepsilon_2^a)^2}{\varpi_2^a} + \frac{1}{2}(\varepsilon_2^a)^2 + \frac{1}{2}(\dot{\alpha}_1^a)^2, \tag{40}$$

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where

$$\dot{\alpha}_1^a = \frac{\partial \alpha_1^a}{\partial \hat{x}_1}\dot{\hat{x}}_1. \tag{41}$$

According to (35)-(40), the virtual control signal α_2^a can be designed as

$$\alpha_2^a = -\beta_2^a \zeta_2^a + \dot{\vartheta}_2^a - \ell_2^a e_1^a - \frac{9}{4}\zeta_2^a, \tag{42}$$

where β_2^a is a positive constant.

In addition, according to (35)-(42), we can get

$$\dot{V}_2^a \le -\beta_2^a(\zeta_2^a)^2 - \frac{1}{4}(\zeta_2^a)^2 + \frac{1}{2}(\zeta_3^a)^2 - \frac{(\varepsilon_2^a)^2}{\varpi_2^a} + \frac{1}{2}(\varepsilon_2^a)^2 + \frac{1}{2}(\varepsilon_3^a)^2 + \frac{1}{2}(\dot{\alpha}_1^a)^2 + \delta_a^3, \tag{43}$$

where $\delta_a^3 = \frac{1}{2}\bar{f}^2 + \frac{1}{2}\Psi_0^2\bar{\kappa}^2$.

Through the above analysis, we select $(1/\varpi_2^a) \ge 3/4 + \gamma_2^a$ with $\gamma_2^a > 0$ being a design parameter. Then, it can be obtained that the following inequality holds:

$$\dot{V}_2^a \le -\beta_2^a(\zeta_2^a)^2 - \frac{1}{4}(\zeta_2^a)^2 + \frac{1}{2}(\zeta_3^a)^2 - (\frac{1}{4} + \gamma_2^a)(\varepsilon_2^a)^2 + \frac{1}{2}(\varepsilon_3^a)^2 + \frac{1}{2}(\dot{\alpha}_1^a)^2 + \delta_a^3. \tag{44}$$

Step j ($3 \le j \le n-1$)*:* Consider the Lyapunov function V_j^a as

$$V_j^a = \frac{1}{2}(\zeta_j^a)^2 + \frac{1}{2}(\varepsilon_j^a)^2. \tag{45}$$

Differentiating V_j^a shows that

$$\dot{V}_j^a = \zeta_j^a(\zeta_{j+1}^a + \alpha_j^a + \varepsilon_{j+1}^a + f_j(q(y^a)) + \hat{\Psi}_a\kappa_j(q(y^a)) + \ell_j^a((q(y^a)) - \hat{y}) - \dot{\vartheta}_j^a) + \varepsilon_j^a\dot{\varepsilon}_j^a. \tag{46}$$

The next steps to be processed are similar to ***Step 2***, we can get

$$\zeta_j^a\zeta_{j+1}^a \le \frac{1}{2}(\zeta_j^a)^2 + \frac{1}{2}(\zeta_{j+1}^a)^2, \tag{47}$$

$$\zeta_j^a\varepsilon_{j+1}^a \le \frac{1}{2}(\zeta_j^a)^2 + \frac{1}{2}(\varepsilon_{j+1}^a)^2, \tag{48}$$

$$\zeta_j^a f_j(q(y^a)) \le \frac{1}{2}(\zeta_j^a)^2 + \frac{1}{2}\bar{f}^2, \tag{49}$$

$$\zeta_j^a\hat{\Psi}_a\kappa_j(q(y^a)) \le \frac{1}{2}(\zeta_j^a)^2 + \frac{1}{2}\Psi_0^2\bar{\kappa}^2, \tag{50}$$

and

$$\varepsilon_j^a\dot{\varepsilon}_j^a \le -\frac{(\varepsilon_j^a)^2}{\varpi_j^a} + \frac{1}{2}(\varepsilon_j^a)^2 + \frac{1}{2}(\dot{\alpha}_{j-1}^a)^2, \tag{51}$$

where

$$\dot{\alpha}_{j-1}^a = \sum_{q=1}^{j-1}\frac{\partial \alpha_{j-1}^a}{\partial \hat{x}_q}\dot{\hat{x}}_q + \sum_{q=2}^{j-1}\frac{\partial \alpha_{j-1}^a}{\partial \vartheta_q^a}\dot{\vartheta}_q^a. \tag{52}$$

The virtual controller α_j^a is chosen as

$$\alpha_j^a = -\beta_j^a\zeta_j^a + \dot{\vartheta}_j^a - \ell_j^a e_1^a - \frac{5}{2}\zeta_j^a, \tag{53}$$

where β_j^a is a positive constant.

From (46) to (53), one obtains

$$\dot{V}_j^a \le -\beta_j^a(\zeta_j^a)^2 - \frac{1}{2}(\zeta_j^a)^2 + \frac{1}{2}(\zeta_{j+1}^a)^2 - \frac{(\varepsilon_j^a)^2}{\varpi_j^a} + \frac{1}{2}(\varepsilon_j^a)^2 + \frac{1}{2}(\varepsilon_{j+1}^a)^2 + \frac{1}{2}(\dot{\alpha}_{j-1}^a)^2 + \delta_a^{j+1}, \tag{54}$$

where $\delta_a^{j+1} = \frac{1}{2}\bar{f}^2 + \frac{1}{2}\Psi_0^2\bar{\kappa}^2$.

Here we choose $(1/\varpi_j^a) \ge 1 + \gamma_j^a$ with $\gamma_j^a > 0$ being the design parameter. Thus, (54) can be rewritten as

$$\dot{V}_j^a \le -\beta_j^a(\zeta_j^a)^2 - \frac{1}{2}(\zeta_j^a)^2 + \frac{1}{2}(\zeta_{j+1}^a)^2 - (\frac{1}{2} + \gamma_j^a)(\varepsilon_j^a)^2 + \frac{1}{2}(\varepsilon_{j+1}^a)^2 + \frac{1}{2}(\dot{\alpha}_{j-1}^a)^2 + \delta_a^{j+1}. \tag{55}$$

Step n: The Lyapunov function V_n^a is chosen as

$$V_n^a = \frac{1}{2}(\zeta_n^a)^2 + \frac{1}{2}(\varepsilon_n^a)^2. \tag{56}$$

Then

$$\dot{V}_n^a = \zeta_n^a(q(u_a) - u_a) + \zeta_n^a(f_n(q(y^a)) + \hat{\Psi}_a\kappa_n(q(y^a)) + \ell_n^a e_1^a - \dot{\vartheta}_n^a + u_a) + \varepsilon_n^a\dot{\varepsilon}_n^a. \tag{57}$$

Combining Young's inequality and (24), the following inequalities always hold:

$$\zeta_n^a(q(u_a) - u_a) \le \frac{c}{4}(\zeta_n^a)^2 + \frac{1}{c}\xi_u^2, \tag{58}$$

$$\zeta_n^a f_n(q(y^a)) \le \frac{1}{2}(\zeta_n^a)^2 + \frac{1}{2}\bar{f}^2, \tag{59}$$

$$\zeta_n^a\hat{\Psi}_a\kappa_n(q(y^a)) \le \frac{1}{2}(\zeta_n^a)^2 + \frac{1}{2}\Psi_0^2\bar{\kappa}^2, \tag{60}$$

and

$$\varepsilon_n^a\dot{\varepsilon}_n^a \le -\frac{(\varepsilon_n^a)^2}{\varpi_n^a} + \frac{1}{2}(\varepsilon_n^a)^2 + \frac{1}{2}(\dot{\alpha}_{n-1}^a)^2, \tag{61}$$

where $c > 0$ is a designed constant,

$$\dot{\alpha}_{n-1}^a = \sum_{q=1}^{n-1}\frac{\partial \alpha_{n-1}^a}{\partial \hat{x}_q}\dot{\hat{x}}_q + \sum_{q=2}^{n-1}\frac{\partial \alpha_{n-1}^a}{\partial \vartheta_q^a}\dot{\vartheta}_q^a. \tag{62}$$

The controller u_a is designed as

$$u_a = -\beta_n^a\zeta_n^a + \dot{\vartheta}_n^a - \ell_n^a e_1^a - (\frac{3}{2} + \frac{c}{4})\zeta_n^a. \tag{63}$$

where β_n^a is a positive constant.

Based on (57) to (62) and selecting $(1/\varpi_n^a) \ge 1 + \gamma_n^a$ with $\gamma_n^a > 0$ being a design parameter, the following inequality can be obtained:

$$\dot{V}_n^a \le -\beta_n^a(\zeta_n^a)^2 - (\frac{1}{2} + \gamma_n^a)(\varepsilon_n^a)^2 + \frac{1}{2}(\dot{\alpha}_{n-1}^a)^2 + \frac{1}{c}\xi_u^2 + \delta_a^{n+1}. \tag{64}$$

where $\delta_a^{n+1} = \frac{1}{2}\bar{f}^2 + \frac{1}{2}\Psi_0^2\bar{\kappa}^2$.

At time $t \in [T_l^a, T_l^b)$, we consider the overall Lyapunov function to be

$$V^a = e^T\mathcal{P}_a e + \sum_{j=1}^{n}\frac{1}{2}(\zeta_j^a)^2 + \sum_{j=2}^{n}\frac{1}{2}(\varepsilon_j^a)^2 + \frac{1}{2}\tilde{\Psi}_a^T\Gamma^{-1}\tilde{\Psi}_a. \tag{65}$$

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According to (19), (33), (44), (55), and (64), we can obtain

$$\begin{aligned}\dot{V}^a \le\ & e^T(\mathcal{P}_a A_1 + A_1^T\mathcal{P}_a + \frac{2}{c} + c\|\mathcal{P}_a\mathfrak{L}_\mathfrak{a}M\|^2 + \frac{4}{c}\|\mathcal{P}_a\|^2\|\mathcal{L}_f^a\|^2 \\ & + \frac{4}{c}\Psi_0^2\|\mathcal{P}_a\|^2\|\mathcal{L}_\kappa^a\|^2 + \tau^a I_1 + \frac{1}{4}I_2)e + \sum_{j=1}^{n-1}\frac{1}{2}(\dot{\alpha}_j^a)^2 + \delta_a \\ & + (4 + \mathcal{L}_{f_1}^a + \hat{\Psi}_a\mathcal{L}_{\kappa_1}^a + \frac{3c}{4} - \frac{\beta_1^a}{2})(\zeta_1^a)^2 - \sum_{j=2}^{n}\beta_j^a(\zeta_j^a)^2 \\ & - \sum_{j=2}^{n}\gamma_j^a(\varepsilon_j^a)^2, \end{aligned} \tag{66}$$

where $\tau^a = (1/2)\beta_1^a$, $\delta_a = \sum_{j=2}^{n}\delta_a^{j+1} + (1/c)\xi_u^2$.

For $\forall\ \Xi_1 > 0$, the set $\Omega_a := \{V^a | V^a = 2e^T\mathcal{P}_a e + \sum_{j=1}^{n}(\zeta_j^a)^2 + \sum_{j=2}^{n}(\varepsilon_j^a)^2 + \tilde{\Psi}_a^T\Gamma^{-1}\tilde{\Psi}_a \le 2\Xi_1\}$ is compact in R^{2n+2}. Further, $|\dot{\alpha}_j^a| \le \Delta_1$ on Ξ_1 for $j = 1, \ldots, n-1$. Then, the following inequality holds:

$$\begin{aligned}\dot{V}^a \le\ & e^T(\mathcal{P}_a A_1 + A_1^T\mathcal{P}_a + \frac{2}{c} + c\|\mathcal{P}_a\mathfrak{L}_\mathfrak{a}M\|^2 + \frac{1}{4}I_2 \\ & + \frac{4}{c}\Psi_0^2\|\mathcal{P}_a\|^2\|\mathcal{L}_\kappa^a\|^2 + \tau^a I_1 + \frac{4}{c}\|\mathcal{P}_a\|^2\|\mathcal{L}_f^a\|^2)e \\ & + (4 + \mathcal{L}_{f_1}^a + \Psi_0\mathcal{L}_{\kappa_1}^a + c + \Psi_m\mathcal{L}_{\kappa_1}^a - \frac{\beta_1^a}{2})(\zeta_1^a)^2 \\ & - \sum_{j=2}^{n}\beta_j^a(\zeta_j^a)^2 - \sum_{j=2}^{n}\gamma_j^a(\varepsilon_j^a)^2 + \bar{\delta}_a, \end{aligned} \tag{67}$$

where $\bar{\delta}_a = \delta_a + (n/2)\Delta_1^2$.

Case 2: When DoS attack is sleeping, for $t \in [T_l^b, T_{l+1}^a)$, the system is not under attack.

Similar to the analysis on ***Case 1***, the actual controller can be designed as

$$u_b = -\beta_n^b\zeta_n^b + \dot{\vartheta}_n^b - \ell_n^b e_1^b - (\frac{3}{2} + \frac{c}{4})\zeta_n^b, \tag{68}$$

where β_n^b is a positive constant. Similar to (67) processing, the following inequality holds:

$$\begin{aligned}\dot{V}^b \le\ & e^T(\mathcal{P}_b A_2 + A_2^T\mathcal{P}_b + \frac{1}{c} + \tau^b I_1 + \frac{1}{4}I_2)e + \bar{\delta}_b \\ & + (4 + \frac{c}{4} - \frac{\beta_1^b}{2})(\zeta_1^b)^2 - \sum_{j=2}^{n}\beta_j^b(\zeta_j^b)^2 - \sum_{j=2}^{n}\gamma_j^b(\varepsilon_j^b)^2, \end{aligned} \tag{69}$$

where $\tau^b = (1/2)\beta_1^b$, $\bar{\delta}_b = \sum_{j=2}^{n}\delta_b^{j+1} + (1/c)\xi_u^2 + (n/2)\Delta_2^2$ with $\Delta_2 \ge |\dot{\alpha}_j^b|$ on $\Omega_b := \{V^b | V^b = 2e^T\mathcal{P}_b e + \sum_{j=1}^{n}(\zeta_j^b)^2 + \sum_{j=2}^{n}(\varepsilon_j^b)^2 + \tilde{\Psi}_b^T\Gamma^{-1}\tilde{\Psi}_b \le 2\Xi_2\}$.

In addition, the estimator design parameter vectors $\mathfrak{L}_\mathfrak{a}$ and $\mathfrak{L}_\mathfrak{b}$ are selected to satisfy the following inequalities:

$$\begin{aligned}& \mathcal{P}_a A_1 + A_1^T\mathcal{P}_a + \frac{2}{c} + c\|\mathcal{P}_a\mathfrak{L}_\mathfrak{a}M\|^2 + \frac{4}{c}\|\mathcal{P}_a\|^2\|\mathcal{L}_f^a\|^2 \\ & + \frac{4}{c}\Psi_0^2\|\mathcal{P}_a\|^2\|\mathcal{L}_\kappa^a\|^2 + \tau^a I_1 + \frac{1}{4}I_2 \le a\mathcal{P}_a \end{aligned} \tag{70}$$

and

$$\mathcal{P}_b A_2 + A_2^T\mathcal{P}_b + \frac{1}{c} + \tau^b I_1 + \frac{1}{4}I_2 \le -b\mathcal{P}_b, \tag{71}$$

where a and b are positive constants.

Furthermore, (67) and (69) can be rewritten as

$$\begin{aligned}\dot{V}^a \le\ & ae^T\mathcal{P}_a e + (4 + \mathcal{L}_{f_1}^a + \Psi_0\mathcal{L}_{\kappa_1}^a + c + \Psi_m\mathcal{L}_{\kappa_1}^a \\ & - \frac{\beta_1^a}{2})(\zeta_1^a)^2 - \sum_{j=2}^{n}\beta_j^a(\zeta_j^a)^2 - \sum_{j=2}^{n}\gamma_j^a(\varepsilon_j^a)^2 + \bar{\delta}_a \end{aligned} \tag{72}$$

and

$$\begin{aligned}\dot{V}^b \le\ & -be^T\mathcal{P}_b e + \bar{\delta}_b + (4 + \frac{c}{4} - \frac{\beta_1^b}{2})(\zeta_1^b)^2 - \sum_{j=2}^{n}\beta_j^b(\zeta_j^b)^2 \\ & - \sum_{j=2}^{n}\gamma_j^b(\varepsilon_j^b)^2. \end{aligned} \tag{73}$$

Meanwhile, the design parameters β_1^a, β_1^b, β_j^a, β_j^b, γ_j^a and γ_j^b $(j = 2, \ldots, n)$ are selected by

$$\begin{cases} -4\big(4 + \mathcal{L}_{f_1}^a + \Psi_0\mathcal{L}_{\kappa_1}^a + c + \Psi_m\mathcal{L}_{\kappa_1}^a - \frac{\beta_1^a}{2}\big) < a \\ -2\beta_j^a < a \\ -2\gamma_j^a < a \end{cases} \tag{74}$$

and

$$\begin{cases} -4\big(4 + \frac{c}{4} - \frac{\beta_1^b}{2}\big) \le -b \\ -2\beta_j^b \le -b \\ -2\gamma_j^b \le -b. \end{cases} \tag{75}$$

According to (72)-(75), the following inequalities hold:

$$\dot{V}^a \le aV^a + \bar{\delta}_a, \tag{76}$$

$$\dot{V}^b \le -bV^b + \bar{\delta}_b. \tag{77}$$

In turn, the derivative of the final Lyapunov function can be expressed as

$$\dot{V} = \begin{cases} \dot{V}^a, & t \in [T_l^a, T_l^b), \\ \dot{V}^b, & t \in [T_l^b, T_{l+1}^a). \end{cases} \tag{78}$$

C. System Stability

Theorem 1: Given the system (9) with input/output quantization under the DoS attacks and satisfying Assumptions 1-3, which include the observer (13), controllers (63), (68) and the parameter law $\hat{\Psi}$. If there exist positive definite matrices $\mathcal{P}_a$, $\mathcal{P}_b$, the vectors $\mathfrak{L}_\mathfrak{a}$, $\mathfrak{L}_\mathfrak{b}$ and constant χ such that they are satisfied

$$\begin{aligned} \mathcal{P}_a &< \hbar\mathcal{P}_a \\ \mathcal{P}_b &< \hbar\mathcal{P}_b \\ \frac{b-\chi}{b+a} &> \frac{1}{\mathcal{T}} \\ \chi &> \frac{\ln\hbar}{\tau_D}. \end{aligned} \tag{79}$$

Then, the designed adaptive output feedback control strategy can guarantee that all signals of the closed-loop system are bounded.

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Proof: Under the conditions (76)-(79), the following inequality holds:

$$\dot{V} \leq \varrho(t)V + \delta, \tag{80}$$

where $\varrho(t) = a$ if $t \in [T_l^a, T_l^b)$, $\varrho(t) = -b$ if $t \in [T_l^b, T_{l+1}^a)$, and $\delta = \bar{\delta}_a + \bar{\delta}_b$. We consider the case $t \in [T_l^a, T_l^b)$, then one has

$$\begin{aligned} V(t) &\leq e^{\varrho_l(t-T_l^a)}V(T_l^{a+}) + \delta\int_{T_l^a}^{t} e^{\varrho_l(t-\pi)}d\pi \\ &\leq \hbar e^{\varrho_l(t-T_l^a)}V(T_l^{a-}) + \delta\int_{T_l^a}^{t} e^{\varrho_l(t-\pi)}d\pi, \end{aligned} \tag{81}$$

where ϱ_l is a constant associated with a or b.

In the lth DoS attack interval, a recursive approach can be employed to obtain

$$\begin{aligned} V(t) \leq\ & \hbar^{n(0,t)}e^{\varrho_l(t-T_l^a)+\cdots+\varrho_0(T_1^a-0)}V(0) \\ &+ \delta\int_0^{T_1^a} \hbar^{n(T_1^a,t)}e^{\varrho_l(t-T_l^a)+\cdots+\varrho_0(T_1^a-\pi)}d\pi \\ &+ \delta\int_{T_1^a}^{T_2^a} \hbar^{n(T_2^a,t)}e^{\varrho_l(t-T_l^a)+\cdots+\varrho_0(T_2^a-\pi)}d\pi \\ &+ \cdots + \delta\int_{T_l^a}^{t} e^{\varrho_l(t-\pi)}d\pi. \end{aligned} \tag{82}$$

Under ***Assumption 1***, we have $|\Theta_a(\pi,t)| \leq (t-\pi/\mathcal{T})$ and $|\Theta_b(\pi,t)| \geq (t-\pi)(1-(1/\mathcal{T}))$ on the time interval $[\pi, t]$. Further, the following inequality can be obtained:

$$\begin{aligned} V(t) \leq\ & \hbar^{n(0,t)}e^{-[b|\Theta_b(0,t)|-a|\Theta_a(0,t)|]}V(0) \\ &+ \delta\int_0^{T_1^a} \hbar^{n(T_1^a,t)}e^{-[b|\Theta_b(\pi,t)|-a|\Theta_a(\pi,t)|]}d\pi \\ &+ \delta\int_{T_1^a}^{T_2^a} \hbar^{n(T_2^a,t)}e^{-[b|\Theta_b(\pi,t)|-a|\Theta_a(\pi,t)|]}d\pi \\ &+ \cdots + \delta\int_{T_l^a}^{t} e^{-[b|\Theta_b(\pi,t)|-a|\Theta_a(\pi,t)|]}d\pi \\ \leq\ & \hbar^{n(0,t)}e^{-[b|\Theta_b(0,t)|-a|\Theta_a(0,t)|]}V(0) \\ &+ \delta\int_0^{t} \hbar^{n(\pi,t)}e^{-[b|\Theta_b(\pi,t)|-a|\Theta_a(\pi,t)|]}d\pi \\ \leq\ & e^{-[b|\Theta_b(0,t)|-a|\Theta_a(0,t)|-n(0,t)\ln\hbar]}V(0) \\ &+ \delta\int_0^{t} e^{-[b|\Theta_b(\pi,t)|-a|\Theta_a(\pi,t)|-n(\pi,t)\ln\hbar]}d\pi. \end{aligned} \tag{83}$$

In the light of ***Assumption 2*** and (79), there exists a constant $\varphi > 0$ satisfies $b|\Theta_b(\pi,t)| - a|\Theta_a(\pi,t)| > \chi(t-\pi) > (n(\pi,t)-\varkappa)\ln\hbar + \varphi(t-\pi)$.

Thus, we have

$$\begin{aligned} V(t) &\leq \hbar^{\varkappa}\left[e^{-\varphi t}V(0) + \delta\int_0^t e^{-\varphi(t-\pi)}d\pi\right] \\ &\leq \hbar^{\varkappa}\left[e^{-\varphi t}V(0) + \frac{\delta}{\varphi}\right]. \end{aligned} \tag{84}$$

Besides, it is easy to obtain that

$$\dot{V} \leq -\varphi V(t) + \delta. \tag{85}$$

Then, this implies that the following inequality holds:

$$V(t) \leq V(0) + \frac{\delta}{\varphi}. \tag{86}$$

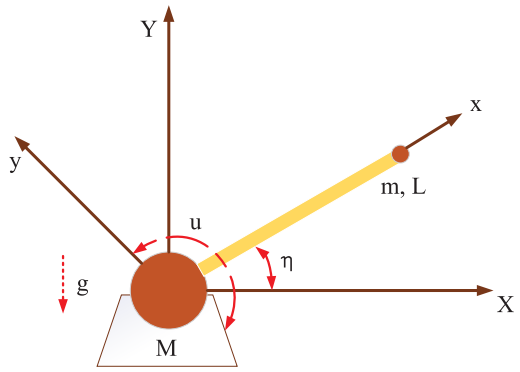


Fig. 2. Framework of single-link robot.

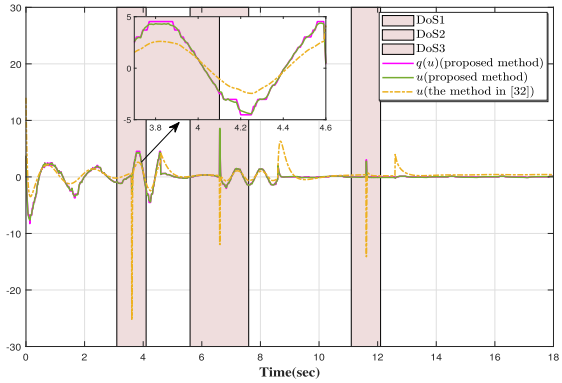


Fig. 3. Curves of $q(u)$ and u.

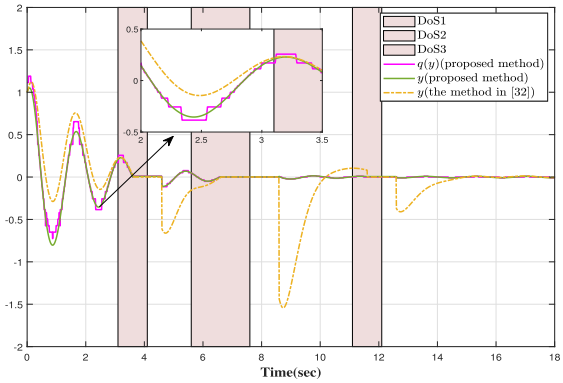


Fig. 4. Curves of $q(y)$ and y.

Based on the above derivations, we can deduce that all signals are bounded and can also ensure that all errors ultimately converge to a small range near the origin. Therefore, we complete the proof. ■

IV. Simulation Studies

In order to verify the reasonableness and effectiveness of the control scheme designed in this paper, we present the following simulation example:

Here, a single-link robot (see Fig.2) borrowed from literature [36] is considered, which can be described in the following form:

$$M\ddot{\eta} + \frac{1}{2}mgL\sin(\eta) = u. \tag{87}$$

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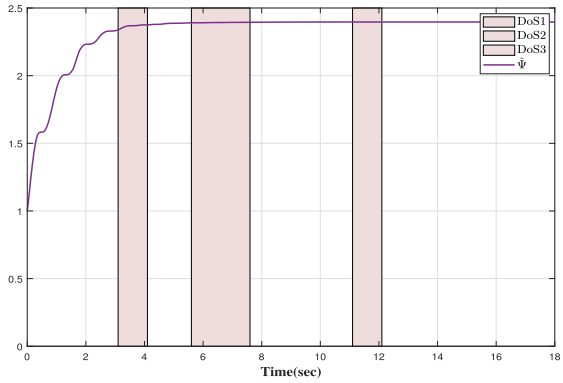


Fig. 5. Curve of $\hat{\Psi}$.

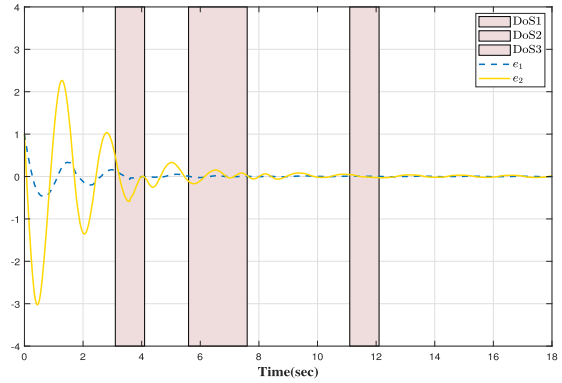


Fig. 6. Curves of e_1 and e_2.

In the simulation, we consider using the logarithmic-uniform quantizer in (3), whose parameters are chosen as $\mu_{lu(y)} = 0.5$, $\mu_{min(y)} = 0.01$, $m_y = 0.2$, $\mu_{lu(u)} = 10$, $\mu_{min(u)} = 0.1$, $m_u = 0.1$. Let $x_1 = \eta$, $x_2 = \dot{\eta}$, $M = 0.5$, $m = 1$, $g = 9.8$, $L = 1$. The initial state is selected as $x_0 = [1; 1]$ and other relevant parameters are selected as $c = 3$, $\varpi_2^a = 10$, $\varpi_2^b = 1$, $\beta_1^a = \beta_2^a = 50$, $\beta_1^b = \beta_2^b = 10$, $\Gamma = 2I_2$, $\mathfrak{L}_\mathfrak{a} = [15, 15]^T$, $\mathfrak{L}_\mathfrak{b} = [5, 5]^T$. The intervals in which the attack occurs can be chosen as $[3.62s, 4.6s] \cup [6.62s, 8.6s] \cup [11.62s, 12.6s]$.

To further demonstrate the advantages of the control method proposed in this paper under DoS attacks, we compared it with a representative control scheme proposed in [32]. Simulations are carried out using the same initial conditions and design parameters with a time step of 0.01s. The above figures show the time required for output as well as input convergence. Compared to the approach proposed in [32], our proposed control strategy not only dominates in terms of performance, but also further reduces the communication burden of the system and improves resource utilization. In order to enhance the practical value of the work in this paper, further practice will be carried out on printed-circuit board [37] as well as microinjection system [38].

V. Conclusion

For the nonlinear system (9) under DoS attacks, this paper proposes an adaptive control strategy based on the observer. By designing a novel QC observer driven by the quantized compensation signal, reconstruction of status signals is achieved. Then, the parameter projection technique is introduced to ensure the reasonableness and effectiveness of parameter estimation. Furthermore, the safety controller is designed using dynamic filtering and backstepping techniques to ensure the boundedness of all closed-loop signals. From the simulation results, the effectiveness of the control scheme is evident. In the practical engineering context, the finite-time adaptive control strategy has the research value of sustainability [39]. Therefore, as for future research directions, our consideration is focused on achieving finite-time adaptive control of multi-input and multi-output (MIMO) nonlinear systems [40] with input/output quantization.

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